

VENKAT ARUN

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EDUCATION

Massachusetts Institute of Technology (MIT) Ph.D. Dept. of EECS Advisors: Hari Balakrishnan and Mohammad Alizadeh	2019-2023
Massachusetts Institute of Technology (MIT) Master of Science, Dept. of EECS Advisors: Hari Balakrishnan and Mohammad Alizadeh	2017-2019
Indian Institute of Technology Guwahati (IIT-G) B.Tech. in Computer Science & Engineering <i>President of India Gold Medal</i>	2013-2017

INDUSTRY IMPACT

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1. Meta uses my congestion control algorithm (CCA), Copa [], for live video uploads
 2. Meta uses a my modification to BBR [] (a CCA designed by Google) for a vast majority of their user-facing traffic

HONORS AND AWARDS

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- MIT EECS G. M. Sprowls PhD Thesis Award in Computer Science (2024)
 - ACM SIGCOMM Doctoral Dissertation Award Runner-Up (2024)
 - Marconi Society Young Scholar Award (2023)
 - ACM SIGCOMM best student paper award (2022)
 - ACM SIGCOMM best paper award (2017)
 - MIT Jacobs Presidential Fellowship (2017)
 - President of India Gold Medal - IIT Guwahati (2017)
 - KVPY Government of India Scholarship (2013)

PUBLICATIONS

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- *On the Role of LLMs in Designing the Architecture of Networked Systems*
Pratyush Sahu, Rahul Bothra, Venkat Arun, Brighten Godfrey, Akshay Narayan, Ahmed Saeed
NSDI 2027
 - *LFS: Learning-Based Flexible CPU Scheduler*
Yeonju Ro, Vincent Zhang, Venkat Arun, Daehyeok Kim, Jiaxin Lin, Christopher J. Rossbach, Atlas Wang, Aditya Akella
NSDI 2027
 - *Synthesizing a Provably Performant Joint Controller for Low-Latency Video Streaming*
Tony Pan, Anup Agarwal, Isil Dillig, Venkat Arun
NSDI 2026
 - *FRCC: Towards Provably Fair and Robust Congestion Control*
Anup Agarwal, Venkat Arun, Srini Seshan
NSDI 2026
 - *Contracts: A unified lens on congestion control robustness, fairness, congestion, and generality*
Anup Agarwal, Venkat Arun, Srinivasan Seshan
NINeS 2026

- *TierSizer: Counterfactual Reasoning for DRAM Sizing in Tiered Memory Systems*
 Sujay Yadalam, Saarth Deshpande, Divyanshu Saxena, Venkat Arun, Neeraja Yadwadkar, Aditya Akella
 DIMES 2026
- *Assistants, Not Architects: The Role of LLMs in Networked Systems Design*
 Pratyush Sahu, Rahul Bothra, Venkat Arun, Brighten Godfrey, Akshay Narayan, Ahmed Saeed
 arXiv 2026
- *Bring Your Own Objective: Inter-operability of Network Objectives in Datacenters*
 Sanjoli Narang, Anup Agarwal, Venkat Arun, Manya Ghobadi
 arXiv 2026
- *Faster-than-light coordination for networked systems with quantum non-local games*
 Venkat Arun, Vijay Chidambaram, Scott Aaronson
 HotNets 2025
- *Robust Heuristic Algorithm Design with LLMs*
 Pantea Karimi, Dany Rouhana, Pooria Namyar, Siva Kesava Reddy Kakarla, Venkat Arun, Behnaz Arzani
 arXiv 2025
- *Speculative Ad-hoc Querying*
 Haoyu Li, Srikanth Kandula, Maria Angels de Luis Balaguer, Aditya Akella, Venkat Arun
 arXiv 2025
- *Lightweight Automated Reasoning for Network Architectures*
 Rahul Bothra, Venkat Arun, Brighten Godfrey, Akshay Narayan, Ahmed Saeed
 HotNets 2024
- *Towards provably performant congestion control*
 Anup Agarwal, Venkat Arun, Devdeep Ray, Ruben Martins, and Srinivasan Seshan
 NSDI 2024
- *Hairpin: Rethinking packet loss recovery in edge-based interactive video streaming*
 Zili Meng, Xiao Kong, Jing Chen, Bo Wang, Mingwei Xu, Rui Han, Honghao Liu, Venkat Arun, Hongxin Hu, and Xue Wei
 NSDI 2024
- *A Performance Verification Methodology for Resource Allocation Heuristics*
 Saksham Goel, Benjamin Mikek, Jehad Aly, Venkat Arun, Ahmed Saeed, and Aditya Akella
 arXiv 2024
- *Starvation in end-to-end congestion control*
 Venkat Arun, Mohammad Alizadeh, and Hari Balakrishnan
 SIGCOMM 2023
Best student paper award
- *Enabling High Quality Real-Time Communications with Adaptive Frame-Rate*
 Zili Meng, Tingfeng Wang, Yixin Shen, Bo Wang, Mingwei Xu, Rui Han, Honghao Liu, Venkat Arun, Hongxin Hu, and Xue Wei
 NSDI 2023
- *Automating network heuristic design and analysis*
 Anup Agarwal, Venkat Arun, Devdeep Ray, Ruben Martins, and Srinivasan Seshan
 HotNets 2022
- *Privid: Practical, Privacy-Preserving Video Analytics Queries*
 Frank Cangialosi, Neil Agarwal, Venkat Arun, Srinivas Narayana, Anand Sarwate, and Ravi Netravali
 NSDI 2022

- *The case for an internet primitive for fault localization*
William Sussman, Emily Marx, Venkat Arun, Akshay Narayan, Mohammad Alizadeh, Hari Balakrishnan, Aurojit Panda, and Scott Shenker
HotNets 2022
- *Toward formally verifying congestion control behavior*
Venkat Arun, Mina T. Arashloo, Ahmed Saeed, Mohammad Alizadeh, and Hari Balakrishnan
SIGCOMM 2021
Proposed a modification to BBR that Meta now uses for most of their production traffic
- *Throughput-fairness tradeoffs in mobility platforms*
Arjun Balasingam, Karthik Gopalakrishnan, Radhika Mittal, Venkat Arun, Ahmed Saeed, Mohammad Alizadeh, Hamsa Balakrishnan, and Hari Balakrishnan
MobiSys 2021
- *RFocus: Beamforming Using Thousands of Passive Antennas*
Venkat Arun and Hari Balakrishnan
NSDI 2020
May be the largest antenna array ever used for a single communication link
- *Finding safety in numbers with secure allegation escrows*
Venkat Arun, Aniket Kate, Deepak Garg, Peter Druschel, and Bobby Bhattacharjee
NDSS 2020
- *Copa: Practical Delay-Based Congestion Control for the Internet*
Venkat Arun and Hari Balakrishnan
NSDI 2018
Being used in production at Meta
- *Language-directed hardware design for network performance monitoring*
Srinivas Narayana, Anirudh Sivaraman, Vikram Nathan, Prateesh Goyal, Venkat Arun, Mohammad Alizadeh, Vimalakumar Jeyakumar, and Changhoon Kim
SIGCOMM 2017
Best paper award

SELECTED PRESS COVERAGE

Starvation in CC	MIT News, <u>IEEE Spectrum</u> , <u>APNIC Blog</u> , The Register, Extreme Tech
RFocus	MIT News, <u>BBC Radio</u> , Tech Crunch, Venture Beat, Engadget, Tech Spot, Digital Trends
Privid	MIT News, <u>IEEE Spectrum</u> , Hacker News, Sci Tech Daily, MarkTechPost